Exercise 1: Core Concepts

a) The core concepts of a case are a problem or situation that need to be solved, a solution for the problem/situation and an outcome/result after the solution was applied.

b) The 4R's are retrieving, reusing, revising and retaining

retrieving: a search from earlier tasks performed that can be of use

reusing: reuse earlier experiences from the retrieving step. May need adjustment and tayloring but the goal is to not need to begin from scratch

revising: this step is about the solution and how it respond on the problem

retaining: storing the new experience so it can be used in a search later

Exercise 2: Details of a CBR System

a) Chosen application: medical diagnosis support system

A system that helps doctors diagnose diseases by comparing a patient's symptoms and medical data with previously stored patient cases. It retrieves similar past cases and suggests diagnoses and treatments that worked before

b) Problem description	Solution	Outcome:
- age - gender - symptoms - body temperature - duration of symptoms	- diagnosis	- used treatment - patient recovery status

Example:

age: 37
gender: male
symptoms: fever, cough, headache
temperature: 38,5°C
blood test: normal values
duration: 3 days
CRP: 50 mg/L
diagnosis: influenza
treatment: rest
outcome: 7 days recovery

c) Symptoms similarity function

$$\text{similarity}(S_1, S_2) = \frac{|S_1 \cap S_2|}{|S_1 \cup S_2|}$$

S_1 is symptoms of patient 1
S_2 is symptoms of patient 2
$S_1 \cap S_2$ symptoms both have
$S_1 \cup S_2$ unique symptoms

Exercise 3)

Knowledge Containers	Relationship - Trade off
Case Base → Vocabulary	Detailed attributes can reduce the number of cases needed
Case Base → Similarity	Better similarity measures can reduce the size of the case base
Case Base → Adaption	More adaptation knowledge allows fewer cases
Similarity → Adaption	Strong similarity reduces adaption effort